Status	Finished
Started	Sunday, 18 October 2026, 11:32 AM
Completed	Sunday, 18 October 2026, 11:40 AM
Duration	7 mins 54 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear

b. Binary ✓

c. Hash

d. Both (1) & (3)

Question **2**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops

b. The higher middle element is chosen as the middle element

c. The middle element is chosen randomly

d. The lower middle element is chosen as the middle element ✗

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓

b. O(n log n)

c. O👎

d. O(1)

Question **4**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array

b. Item is the last element in the array or item is not there at all

c. When the item is somewhere in the middle of the array ✓

d. When the item is not the array at all

Question **5**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list

b. Deleting elements from a list

c. Determining whether an element is present in a list ✓

d. Inserting elements into a list

Question **6**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n

b. n ✓

c. 1

d. n/2

Question **7**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining
b. Searching ✓
c. Finding
d. Discovering

Question **8**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the list is sorted
c. When the element is found in the list ✓
d. When the search algorithm finishes

Question **9**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search backtracks to find duplicate elements
c. The search stops immediately ✓
d. The search starts over from the beginning

Question **10**

Incorrect

Mark 0.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list

b. There should be direct access to the middle element in any sublist

c. Number values should only be present ✗

Number values should only be present

d. The list must be sorted

Question **11**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list

b. Binary search has a lower time complexity on large, sorted lists ✓

c. Binary search works on unsorted lists

d. Binary search can find multiple instances of the target element

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. Must use a sorted array

Question **13**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search continues in the right sublist
c. The search ends successfully ✓
d. The search continues in the left sublist

Question **14**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Sequential search
c. Binary search ✓
d. Hash search

Question **15**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Jump search
c. Linear search ✓
d. Binary search